Bangladesh University of Engineering and Technology

Course: CSE 316

Sections: A1, B1

Problem Statement:

Write an 8086 assembly language program (compatible with the emu8086 emulator) to find the n-th term of the Fibonacci series. Your program must satisfy the following technical requirements:

1. **Multiple Digit Input:** The program must prompt the user to enter a multi-digit decimal number n, where $0 \leq n \leq 24$.
2. **Computation:** Calculate the n-th Fibonacci number, $F(n)$, based on the standard recurrence relation:

$$\begin{aligned} F(0) &= 0 \\ F(1) &= 1 \\ F(n) &= F(n-1) + F(n-2) \quad \text{for } n \geq 2 \end{aligned}$$

3. **Multiple Digit Output:** The program must convert the 16-bit binary result back into decimal and print the multi-digit Fibonacci number to the console.

Note: The constraint $n \leq 24$ is given because $F(24) = 46368$, which is the largest Fibonacci number that can be stored in a standard 16-bit unsigned register (maximum value 65535) without causing an overflow.